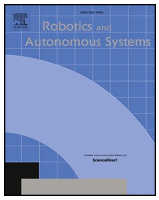
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A novel digital twin model based bio-heuristic sliding mode control algorithm for trajectory tracking control of USV in the presence of complex marine environment disturbance

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ABSTRACT

The trajectory tracking control problem of unmanned surface vessel (USV) under the complex marine environmental disturbance is discussed in this paper, and a novel digital twin model based bio-heuristic sliding mode control (SMC) algorithm integrated with a radial basis function neural network (RBFNN) disturbance compensation module is proposed. An adaptive forgetting factor, which varies with the prediction errors of state variables, is introduced and integrated into the recursive least squares (RLS) algorithm. Meanwhile, a digital twin model of USV is established by applying the proposed adaptive forgetting factor recursive least squares (AFF-RLS) algorithm, and utilizing state variable data and control commands. An improved bio-heuristic approximation function is presented to approach the virtual velocity control laws, avoiding abrupt change and jittering of the SMC algorithm designed based on the digital twin model. Set the angular and linear velocity variables as inputs, environment disturbances and modeling errors compensation as outputs, the RBFNN based integrated control compensator is presented, where the minimum parameter learning method is implemented by replacing the adjustment of neural network weights with parameter estimation to reduce redundant parameters. The effectiveness of the proposed algorithm is validated through extensive simulation experiments, demonstrating its robustness in real marine environment disturbance.

1. Introduction

In recent years, the unmanned surface vehicle (USV) has garnered increasing attention due to its multifunctionality and adaptability in the context of increasing global emphasis on marine development. USV has significant application value, it could effectively collect marine data to support real-time assessments and enhance operational safety and flexibility in areas such as ocean monitoring, disaster response, cargo transportation, scientific research and environmental protection [1,2]. The trajectory tracking control of USV is essential for realizing its capability, serving as a vital foundation for its application in various aspects. Over the past few decades, the trajectory tracking control problem of underactuated USV has emerged as a significant research focus [3].

In the research field of trajectory tracking control for USV, parametric uncertainties and external marine environmental disturbances could adversely affect tracking performance [4]. Many scholars have conducted extensive research on the trajectory tracking control algorithms of USV to improve its tracking performance, including sliding mode control (SMC) [5,6], model predictive control (MPC) [7,8] and backstepping control [9]. Among these algorithms, the SMC algorithm is widely utilized in industrial control, USV, aircraft and other fields due to its robustness and fast response. A novel kind of terminal sliding mode control (TSMC) algorithm is proposed by Zhang in [10] to enhance the robustness of the unmanned surface vehicle-remotely operated vehicles (USV-ROVs). The adaptive control law is designed in literature [11] based on the sliding mode function to improve USV's tracking performance under the complex external environment. Liu in [12] proposed a hierarchical sliding mode control strategy to address the formation-containment control problem of multiple underactuated

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USVs. In [13], the nonfragile SMC algorithm is presented to enhance USV's trajectory tracking control performance under environmental disturbances.

An observer-based fixed-time tracking control strategy is presented in [14] for USVs with model uncertainties, external disturbances and actuator faults. To address the complex uncertain hydrodynamics and system constraints, a novel kind of hybrid physics-learning model based predictive control (PL-MPC) algorithm is proposed by Zheng in [15]. And [16] presented an improved guidance law based on the training characteristics of deep reinforcement learning (DRL) algorithms to achieve accurate trajectory tracking of USV. A robust trajectory tracking controller is designed by [17] to overcome model uncertainties, unmeasurable disturbance and noise. In order to solve the guaranteed trajectory estimation problem for USV, an offline data-based set-membership estimation algorithm of unknown system parameters is proposed to obtain the system parameters by [18]. In [19], the remote estimation of states and model parameters for a class of USV with unknown noise parameters is addressed by develop a new class of online estimation approach based on the variational Bayesian technique. A new kind of adaptive unscented Kalman filter is proposed in [20] to deal with the parameter estimation problem for a class of USV models with unknown process noises. A stochastic gradient algorithm with a forgetting factor is presented by [21] to realize parameter estimation.

In addition, an augmented model-based finite-time observer (AM-FTO) is proposed by [22] to observe the environmental disturbances. A prescribed-time extended state observer (PTESO) guidance law is presented by [23] for the accurate determination of the sideslip angle of the USV, particularly during maneuvering along curved trajectories or in the presence of time-varying marine disturbances. To achieve control of the USV at a higher level, a model-free finite-time trajectory-tracking control (MFTC) scheme, incorporating a finite-time position observer (FPO) and a finite-time extended state observer (FESO), is designed in [24].

Through the analysis of the above literature, it is easy to find that model-based control algorithms are widely applied in the field of USV trajectory tracking control as shown in [10–13]. The fixed model parameters utilized in these algorithms are usually obtained by computational fluid dynamics (CFD) software calculations [25], constrained model experiments, and empirical formulas [26], and there are certain modeling errors, which can reduce the performance of the designed controllers. Online parameter identification modeling is a novel approach to obtain model parameters of the USV, the identification algorithms in the existing literature [18–21] are at risk of data saturation, the implementation of fixed weight values for historical data could result in the new data that cannot correct the identification results.

Simultaneously, SMC algorithms are extensively utilized in trajectory tracking control of USV, the output mutation and jitter during the controller design process often existed, and meanwhile the defects of computational inefficiency are found in papers [10–13], both of them may weaken the performance of the SMC algorithm. Disturbance observers designed in literature [22–24] heavily rely on model accuracy, and they cannot adapt effectively to unknown or rapidly changing disturbances.

Motivated by the above analysis, a novel digital twin model based bio-heuristic SMC algorithm, integrated with an RBFNN compensator, is proposed for trajectory tracking of an underactuated USV. A digital twin model of the USV is constructed by applying the online identification algorithm, which updates model parameters in real time based on state data and control outputs. By adjusting forgetting factor change interval and designed prediction error value, the adaptive forgetting factor recursive least square (AFF-RLS) algorithm is proposed based on real-time prediction error data to adaptively adjust the forgetting factor. An enhanced bio-heuristic approximation function is presented to approximate virtual velocity control variables, smoothing the trajectory tracking control variables and then preventing the sudden changes and oscillations. Additionally, a compensator based on RBFNN is proposed to counteract disturbances and digital twin modeling errors, where the angular and linear velocities are inputs, and compensations for disturbances and modeling errors are outputs. In comparison to previous studies, the key contributions of this research are outlined as follows:

(1) A digital twin model of the USV is constructed by integrating online identification algorithm, enabling real-time updates of model parameters based on state data and control outputs, thereby enhancing model accuracy and robustness in complex marine environment.

(2) The forgetting factors in the AFF-RLS algorithm is adaptively adjusted based on real-time prediction error data, with intervals and designed prediction errors set to optimize parameter estimation accuracy.

(3) An enhanced bio-heuristic approximation function is designed within the sliding mode controller to approximate the virtual velocity control input, smoothing the trajectory tracking control and reducing chattering, thereby preventing sudden changes and oscillations to improve tracking performance.

(4) A RBFNN-based compensator is proposed, where angular and linear velocities serve as inputs, disturbances and modeling errors are compensated for, with the weight vector transformed into a single-parameter form to simplify calculations and improve disturbance rejection.

The rest of this paper is structured as follows: the trajectory tracking control problem of the USV is introduced in Section 2. In Section 3, the trajectory tracking controller is outlined. A composite stability proof for the controller and the RBFNN algorithm is presented in Section 4. Several simulation experiments are conducted in Section 5. Finally, Section 6 concludes this paper.

2. Problem description

Trajectory tracking control problem of the USV, which refers to designing the control law to force the USV tracking the desired trajectory, is discussed in this paper. It could be shown as follow:

As presented in Fig.1, the USV's trajectory tracking control problem mainly includes two key components: the establishment of the USV model and the formulation of the trajectory tracking control, which could be described as follows:

2.1. The establishment of USV's mathematical model

According to Fig.1, the kinematic model of the USV in the horizontal plane could be expressed as follows:

$$\begin{cases} \dot{x} = u\cos\psi - v\sin\psi \\ \dot{y} = u\sin\psi + v\cos\psi \\ \dot{\psi} = r \end{cases} \tag{1}$$

where $x(m), y(m), \psi(rad)$ represent the longitudinal position, lateral position and heading angle of the USV; $u(m/s), v(m/s), r(rad/s)$ are the velocities in the longitudinal, lateral and yaw directions of the USV; Assuming that the USV is symmetric with respect to the longitudinal and transverse sections, and only the first-order linear damping term of the USV is considered. Based on the Fossen model presented in [27], the dynamic model of the underactuated USV could be expressed as follows:

$$\mathbf{M}\dot{\boldsymbol{v}} + \mathbf{C}(\boldsymbol{v})\boldsymbol{v} + \mathbf{D}(\boldsymbol{v})\boldsymbol{v} = \boldsymbol{\tau} + \mathbf{d}_o \tag{2}$$

where $\boldsymbol{v} = [u, v, r]^T$; τ represents the longitudinal thrust and turning moment of USV, $\boldsymbol{\tau} = [\tau_u, 0, \tau_r]^T$; $\mathbf{d}_0$ represents the disturbance forces and moments caused by marine disturbance in the longitudinal, lateral, and yaw directions, $\mathbf{d}_o = [d_{ou}, d_{ov}, d_{or}]^T$; $\mathbf{M}$ is the inertia matrix of the USV; $\mathbf{C}(\boldsymbol{v})$ is the Coriolis-centripetal force matrix of the USV; $\mathbf{D}(\boldsymbol{v})$ is the damping coefficient matrix; The expressions of each matrix are shown as :

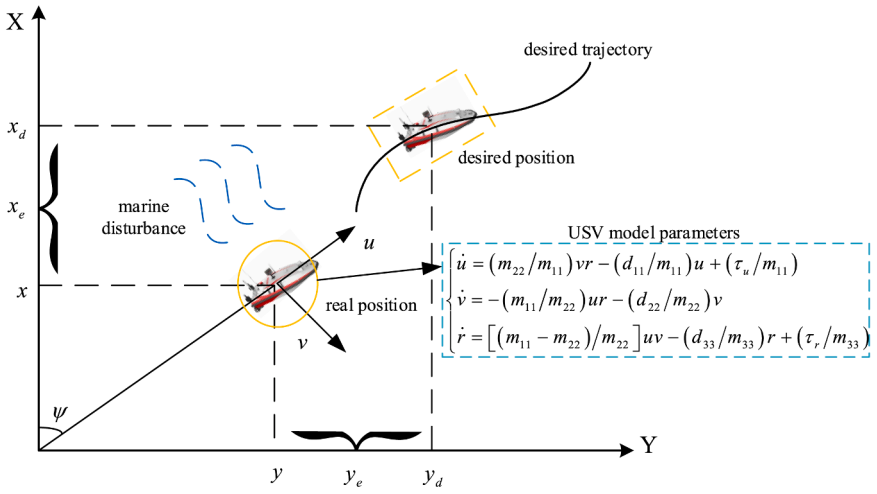


Fig. 1. Schematic diagram of USV's trajectory tracking.

$$\begin{cases} \mathbf{M} = \begin{bmatrix} m_{11} & 0 & 0 \\ 0 & m_{22} & 0 \\ 0 & 0 & m_{33} \end{bmatrix}; \mathbf{D}(\boldsymbol{\upsilon}) = \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix} \\ \mathbf{C}(\boldsymbol{\upsilon}) = \begin{bmatrix} 0 & 0 & c_{13} \\ 0 & 0 & c_{23} \\ c_{31} & c_{32} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -m_{22}v \\ 0 & 0 & m_{11}u \\ m_{22}v & -m_{11}u & 0 \end{bmatrix} \end{cases} \tag{3}$$

where $m_{11}, m_{22}, m_{33}, d_{11}, d_{22}, d_{33}$ are model parameters to be identified. According to Eq. (1), Eq. (2) and Eq. (3), the Fossen model of USV could be obtained as follows:

$$\begin{cases} \dot{x} = u\cos\psi - v\sin\psi \\ \dot{y} = u\sin\psi + v\cos\psi \\ \dot{\psi} = r \\ \dot{u} = m_{22}vr/m_{11} - d_{11}u/m_{11} + (\tau_u + d_{ou})/m_{11} \\ \dot{v} = -m_{11}ur/m_{22} - d_{22}v/m_{22} + d_{ov}/m_{22} \\ \dot{r} = (m_{11} - m_{22})uv/m_{22} - d_{33}r/m_{33} + (\tau_r + d_{or})/m_{33} \end{cases} \tag{4}$$

2.2. The formulation of the trajectory tracking control

As shown in Fig.1, during the process of USV's trajectory tracking, the deviations in longitudinal and lateral directions between the real and desired positions could be written as:

$$\begin{cases} x_e = x - x_d \\ y_e = y - y_d \end{cases} \tag{5}$$

where (x, y) is the real position of the underactuated USV in the horizontal plane; (x_d, y_d) is the desired position. The trajectory tracking problem of underactuated USV could be formulated as the design of appropriate control outputs τ_u and τ_r, ensuring that the position deviations x_e and y_e converge to zero.

3. Digital twin model design

As reported by [28], Digital twins are dynamic virtual representations of physical systems that enable real-time data synchronization across the cycle. In the digital twin model design, a linearized identification model is constructed by rearranging terms of the Fossen model, establishing a dynamic mapping between state data and control commands. Inspired by the real-time compensation mechanism, the AFF-RLS algorithm dynamically adjusts the gain matrix to minimize the prediction error based on current inputs, enabling online optimization of the parameter matrix. These optimized parameters directly drive the construction and updating of the USV digital twin model, forming a virtual representation system that integrates embedded physical laws and robustness against environmental disturbances. The specific design process is shown in Fig.2:

As shown in Fig.2, the design process of the digital twin model includes two components: the establishment of the identification model, which demonstrates the mapping relationship between input and output by separating the parameters to be identified; and the construction of the digital twin model based on the AFF-RLS algorithm, which updates the USV model parameters in real-time and establishes the digital twin based on these parameters, ultimately achieving closed-loop adaptive synchronization between physical and virtual entities. The detailed description is as follows:

3.1. Design of the identification model

During the digital twin modeling process, as reported by [29], parameter identification relies on the development of an identification model that accurately reflects the input-output relationship of the USV. To simplify the analysis, the mathematical model is designed in a linearized form that represents the system's input-output relationship, as expressed below:

$$\mathbf{Y} = \mathbf{H}\theta \tag{6}$$

where $\mathbf{Y}$ is the output matrix; $\mathbf{H}$ is the input matrix; $\boldsymbol{\theta}$ is the parameter matrix.

To simplify the longitudinal motion equation and isolate the parameters to be identified, the fourth equation in Eq. (4) is rewritten as follows:

$$\begin{aligned} \dot{u} &= m_{22}vr/m_{11} - d_{11}u/m_{11} + (\tau_u + d_{ou})/m_{11} \\ &= (m_{22}/m_{11})vr - (d_{11}/m_{11})u + (1/m_{11})(\tau_u + d_{ou}) \end{aligned} \tag{7}$$

To transform Eq. (7) into Eq. (6), the following matrix could be designed:

$$\begin{cases} \mathbf{Y}_u = [\dot{u}] \\ \mathbf{H}_u = [vr, u, \tau_u + d_{ou}]^T \\ \boldsymbol{\theta}_u = [m_{22}/m_{11}, -d_{11}/m_{11}, 1/m_{11}] \end{cases} \tag{8}$$

where $\mathbf{Y}_u, \mathbf{H}_u$ could be obtained based on Eq. (7); $\boldsymbol{\theta}_u = [m_{22}/m_{11}, -d_{11}/m_{11}, 1/m_{11}]$ are the model parameters to be identified in the longitudinal direction, and then Eq. (7) could be expressed as follows:

$$\mathbf{Y}_u = \mathbf{H}_u\boldsymbol{\theta}_u \tag{9}$$

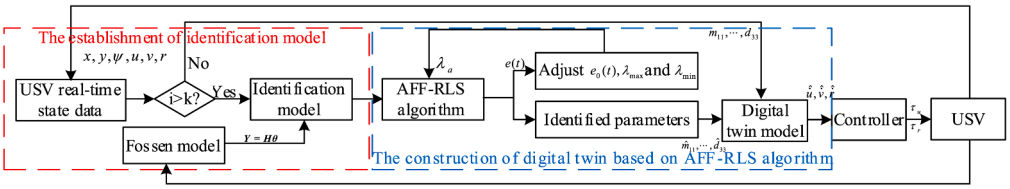


Fig. 2. The digital twin model design process.

Similarly, to isolate the identification of parameters in the lateral direction, the fifth equation in Eq. (4) is expressed to simplify the motion equation as follows:

$$\begin{aligned}\dot{v} &= -m_{11}ur/m_{22} - d_{22}v/m_{22} + d_{ov}/m_{22} \\ &= -(m_{11}/m_{22})ur - (d_{22}/m_{22})v + (1/m_{22})d_{ov}\end{aligned} \tag{10}$$

To transform Eq. (10) into Eq. (6), the following matrix could be designed:

$$\begin{cases} \mathbf{Y}_v = [\dot{v}] \\ \mathbf{H}_v = [ur, v, d_{ov}]^T \\ \boldsymbol{\theta}_v = [-m_{11}/m_{22}, -d_{22}/m_{22}, 1/m_{22}] \end{cases} \tag{11}$$

where $\mathbf{Y}_v, \mathbf{H}_v$ could be obtained based on Eq. (7); $\boldsymbol{\theta}_v = [-m_{11}/m_{22}, -d_{22}/m_{22}, 1/m_{22}]$are the model parameters to be identified in the lateral direction, and then Eq. (10) could be expressed as follows:

$$\mathbf{Y}_v = \mathbf{H}_v\boldsymbol{\theta}_v \tag{12}$$

For the purpose of simplifying the motion equation and isolating the model parameters in yaw direction, the sixth equation in Eq. (4) could be restructured in the following form:

$$\begin{aligned}\dot{r} &= (m_{11} - m_{22})uv/m_{22} - d_{33}r/m_{33} + (\tau_r + d_{or})/m_{33} \\ &= [(m_{11} - m_{22})/m_{22}]uv - (d_{33}/m_{33})r + (1/m_{33})(\tau_r + d_{or})\end{aligned} \tag{13}$$

To transform Eq. (13) into Eq. (6), the following matrix could be designed:

$$\begin{cases} \mathbf{Y}_r = [\dot{r}] \\ \mathbf{H}_r = [uv, r, \tau_r + d_{or}]^T \\ \boldsymbol{\theta}_r = [(m_{11} - m_{22})/m_{22}, (d_{33}/m_{33}), (1/m_{33})] \end{cases} \tag{14}$$

where$\mathbf{Y}_r$, $\mathbf{H}_r$could be obtained based on Eq. (7); $\boldsymbol{\theta}_r = [(m_{11} - m_{22})/m_{22}, (d_{33}/m_{33}), (1/m_{33})]$ are the model parameters to be identified in the lateral direction, and then Eq. (13) could be expressed as follows:

$$\mathbf{Y}_r = \mathbf{H}_r\boldsymbol{\theta}_r \tag{15}$$

The mathematical model of the USV has been standardized into a linearized identification form that reflects the USV's input-output relationship, making it suitable for application by a parameter identification algorithm, which could be expressed as:

$$\begin{cases} \mathbf{Y}_u = \mathrm{H}_u\boldsymbol{\theta}_u \\ \mathbf{Y}_v = \mathrm{H}_v\boldsymbol{\theta}_v \\ \mathbf{Y}_r = \mathrm{H}_r\boldsymbol{\theta}_r \end{cases} \tag{16}$$

3.2. The construction of the digital twin model based on the AFF-RLS algorithm

The recursive least squares (RLS) algorithm is an online parameter identification method based on filtering theory. It offers several advantages, including ease of implementation, high accuracy, and low computational complexity, which contribute to its wide applicability. The general structure of the RLS algorithm is outlined below [30]:

$$\begin{cases} \widehat{\boldsymbol{\theta}}(t) = \widehat{\boldsymbol{\theta}}(t-1) + \boldsymbol{K}(t)[\boldsymbol{Y}(t) - \boldsymbol{H}(t)\widehat{\boldsymbol{\theta}}(t-1)] \\ \boldsymbol{K}(t) = \boldsymbol{P}(t-1)\boldsymbol{H}(t)^T[\boldsymbol{I} + \boldsymbol{H}(t)\boldsymbol{P}(t-1)\boldsymbol{H}(t)^T]^{-1} \\ \boldsymbol{P}(t) = [\boldsymbol{I} - \boldsymbol{K}(t)\boldsymbol{H}(t)]\boldsymbol{P}(t-1) \\ \mathbf{e}(t) = \mathbf{Y}(t) - \mathbf{H}(t)\widehat{\theta}(t-1) \end{cases} \tag{17}$$

where $\boldsymbol{K}(t)$ is the gain matrix; $\boldsymbol{P}(t)$ is the covariance matrix; $\boldsymbol{H}(t)$ is an input matrix; $\boldsymbol{I}$ is the unit matrix; $\widehat{\boldsymbol{\theta}}(t)$ is the identified parameters matrix; $\boldsymbol{Y}(t)$ is the actual output matrix; $\mathbf{e}(t)$ is the prediction error matrix.

A potential limitation inherent in the standard RLS algorithm as formulated in Eq. (17) is that as more data is processed, the covariance matrix $\boldsymbol{P}(t)$ tends to diminish asymptotically towards zero. This leads to a corresponding decay in the gain vector $\boldsymbol{K}(t)$, resulting in a diminished responsiveness of the algorithm to new incoming data. While this reduced sensitivity is a natural and desired characteristic of the algorithm after the parameters have fully converged to their true constant values, indicating stability, it poses challenges during the convergence phase. A prematurely reduced gain $\boldsymbol{K}(t)$ can impede the algorithm's ability to utilize new data effectively to correct estimation errors in the initial learning period, potentially prolonging convergence time. Furthermore, the monotonic decay of $\boldsymbol{P}(t)$ can lead to numerical ill-conditioning or singularity in practical implementations with finite precision arithmetic, hindering stable adaptation or even halting updates. To address these challenges during the convergence process, particularly concerning accelerating the initial convergence and maintaining the numerical robustness of the covariance matrix, this paper employs a forgetting factor recursive least squares (FF-RLS) algorithm. The FF-RLS mechanism introduces a forgetting factor ($\lambda, \lambda < 1$) that selectively discounts the influence of older data points by exponentially weighting them. By doing so, it helps preserve the boundedness of the $\boldsymbol{P}(t)$ matrix and sustains a non-vanishing gain $\boldsymbol{K}(t)$, thereby enhancing the convergence speed and numerical stability during the learning transient. The identification structure of the FF-RLS algorithm is outlined as follows:

$$\begin{cases} \widehat{\boldsymbol{\theta}}(t) = \widehat{\boldsymbol{\theta}}(t-1) + \boldsymbol{K}(t)\mathbf{e}(t) \\ \boldsymbol{K}(t) = \boldsymbol{P}(t-1)\boldsymbol{H}(t)^T[\lambda + \boldsymbol{H}(t)\boldsymbol{P}(t-1)\boldsymbol{H}(t)^T]^{-1} \\ \boldsymbol{P}(t) = [\boldsymbol{I} - \boldsymbol{K}(t)\boldsymbol{H}(t)]\boldsymbol{P}(t-1)/\lambda \\ \mathbf{e}(t) = \mathbf{Y}(t) - \mathbf{H}(t)\widehat{\theta}(t-1) \end{cases} \tag{18}$$

where $\boldsymbol{\lambda}$ is the forgetting factor matrix. The value of the forgetting factor λ significantly affects the accuracy of the FF-RLS algorithm. A large forgetting factor amplifies the influence of older data, reducing the algorithm's sensitivity to new information and thereby impairing its adaptability and real-time performance. In contrast, a small forgetting factor leads to performance instability and a slower convergence rate.

To overcome the challenge that the fixed forgetting factor cannot adaptively adjust with changes in the prediction error, an AFF-RLS algorithm is proposed in this paper, which dynamically adjusts the forgetting factor based on the designed prediction error and change interval. When the prediction error is large, the forgetting factor is quickly reduced to its minimum value, thereby improving identification accuracy. When the error falls within an acceptable range, the forgetting factor increase towards its maximum value, which enhance the stability of parameter identification. The adaptive forgetting factor could be designed as follows:

$$\begin{cases} \mu(t) = round(|e(t)/e_0|^n) \\ \lambda_a(t) = \lambda_{\max} - (\lambda_{\max} - \lambda_{\min})\arctan[\mu(t)]\cdot 2/\pi \end{cases} \tag{19}$$

where, λ_a is the adaptive forgetting factor; $\lambda_{\max}$ is the designed maximum forgetting factor; $\lambda_{\min}$ is the designed minimum forgetting factor; $e(t)$ is the prediction error; When $e(t) > e_0$, the ratio will be decreased to accelerate the λ_a converge to the designed maximum forgetting factor; In contrast, when $e(t) < e_0$, the ratio will be increased further to accelerate the λ_a converge to the designed minimum forgetting factor. The integer function enables gradient-based adjustments for numerical matching, thereby reducing intermediate transition values.

Considering Eq. (19), the expression of the AFF-RLS could be designed as:

$$\begin{cases} \widehat{\boldsymbol{\theta}}(t) = \widehat{\boldsymbol{\theta}}(t-1) + \boldsymbol{K}(t)\mathbf{e}(t) \\ \boldsymbol{K}(t) = \boldsymbol{P}(t-1)\boldsymbol{H}(t)^T\left[\lambda_a + \boldsymbol{H}(t)\boldsymbol{P}(t-1)\boldsymbol{H}(t)^T\right]^{-1} \\ \boldsymbol{P}(t) = [\boldsymbol{I} - \boldsymbol{K}(t)\boldsymbol{H}(t)]\boldsymbol{P}(t-1)/\lambda_a \\ \mathbf{e}(t) = \mathbf{Y}(t) - \mathbf{H}(t)\widehat{\theta}(t-1) \end{cases} \tag{20}$$

where λ_a is the adaptive forgetting factor; $\widehat{\theta} = diag([\widehat{\theta}_u, \widehat{\theta}_v, \widehat{\theta}_r])$, which could be expressed as follows:

$$\begin{cases} \widehat{\theta}_u = [\widehat{m}_{22}/\widehat{m}_{11}, -\widehat{d}_{11}/\widehat{m}_{11}, 1/\widehat{m}_{11}] \\ \widehat{\theta}_v = [-\widehat{m}_{11}/\widehat{m}_{22}, -\widehat{d}_{22}/\widehat{m}_{22}, 1/\widehat{m}_{22}] \\ \widehat{\theta}_r = [(\widehat{m}_{11} - \widehat{m}_{22})/\widehat{m}_{22}, \widehat{d}_{33}/\widehat{m}_{33}, 1/\widehat{m}_{33}] \end{cases} \tag{21}$$

where-
$\widehat{m}_{22}/\widehat{m}_{11}, -\widehat{d}_{11}/\widehat{m}_{11}, 1/\widehat{m}_{11}, -\widehat{m}_{11}/\widehat{m}_{22}, -\widehat{d}_{22}/\widehat{m}_{22}, 1/\widehat{m}_{22}, (\widehat{m}_{11} - -\widehat{m}_{22})/\widehat{m}_{22}, \widehat{d}_{33}/\widehat{m}_{33}, 1/\widehat{m}_{33}$ are dimensional form of the identified model parameters.

Based on Eq. (4) and Eq. (21), the digital twin model of the USV could be constructed as:

$$\begin{cases} \dot{u} = \widehat{m}_{22}vr/\widehat{m}_{11} - \widehat{d}_{11}u/\widehat{m}_{11} + (\tau_u + d_{ou} + d_{op})/\widehat{m}_{11} \\ \quad = \widehat{m}_{22}vr/\widehat{m}_{11} - \widehat{d}_{11}u/\widehat{m}_{11} + (\tau_u + d_u)/\widehat{m}_{11} \\ \quad = \widehat{m}_{22}vr/\widehat{m}_{11} - \widehat{d}_{11}u/\widehat{m}_{11} + \tau_u^*/\widehat{m}_{11} \\ \dot{v} = -\widehat{m}_{11}ur/\widehat{m}_{22} - \widehat{d}_{22}v/\widehat{m}_{22} + (d_{ov} + d_{pv})/\widehat{m}_{22} \\ \quad = -\widehat{m}_{11}ur/\widehat{m}_{22} - \widehat{d}_{22}v/\widehat{m}_{22} + d_v/\widehat{m}_{22} \\ \dot{r} = (\widehat{m}_{11} - \widehat{m}_{22})uv/\widehat{m}_{22} - \widehat{d}_{33}r/\widehat{m}_{33} + (\tau_r + d_{or} + d_{pr})/\widehat{m}_{33} \\ \quad = (\widehat{m}_{11} - \widehat{m}_{22})uv/\widehat{m}_{22} - \widehat{d}_{33}r/\widehat{m}_{33} + (\tau_r + d_r)/\widehat{m}_{33} \\ \quad = (\widehat{m}_{11} - \widehat{m}_{22})uv/\widehat{m}_{22} - \widehat{d}_{33}r/\widehat{m}_{33} + \tau_r^*/\widehat{m}_{33} \end{cases} \tag{22}$$

where d_u represent the composite disturbance in the longitudinal direction, which consists of marine disturbance $d_{ou}$and identification modeling errord_{pu}, $\tau_u^* = \tau_u + d_{ou} + d_{pu}$; d_v represent the composite disturbance in the lateral direction, which consists of marine disturbance d_{ov} and identification modeling error d_{pv}; d_r represent the composite disturbance in the yaw direction, which consists of marine disturbance d_{ro} and identification modeling error d_{rp}; $\tau_r^* = \tau_r + d_{or} + d_{pr}$.

4. The controller design based on the digital twin model

The controller design based on the digital twin model is summarized as: A virtual velocity control law is proposed to ensure the position errors converge to zero, then a bio-heuristic model is applied to design an approximation function for this law, and the sign function is replaced with a hyperbolic tangent function to avoid complex derivations and chattering phenomena; Furthermore, an RBFNN-based disturbance compensation module is presented, with linear and angular velocity as inputs, and the observation values as outputs, the interference suppression capability of the sliding mode control law is further optimized by converting the weight vector into a single-parameter form, which is then applied to the controller design. The detailed design process is shown in Fig.3:

As shown in Fig.3, the controller design process based on digital twin model includes two components: the design of the bio-heuristic SMC algorithm, and the establishment of the RBFNN disturbance observation module. The detailed description is as follows:

4.1. The design of the bio-heuristic SMC algorithm

According to the expressions for longitudinal and lateral deviations in Eq. (5), the derivative of Eq. (5) could be written as follows:

$$\begin{cases} \dot{x}_e = \dot{x} - \dot{x}_d = u\cos\psi - v\sin\psi - \dot{x}_d \\ \dot{y}_e = \dot{y} - \dot{y}_d = u\sin\psi + v\cos\psi - \dot{y}_d \end{cases} \tag{23}$$

To ensure stable convergence of trajectory tracking errors, the virtual design principle of the nonlinear backstepping control is incorporated, with longitudinal and lateral velocities serving as virtual control variables. The virtual control laws could be designed as follows:

$$\begin{cases} \alpha_u = \left[\dot{x}_d - (k_1 x_e)\Big/\sqrt{x_e^2 + y_e^2 + C}\right]\cos\psi + \left[\dot{y}_d - (k_2 y_e)\Big/\sqrt{x_e^2 + y_e^2 + C}\right]\sin\psi \\ \alpha_v = -\left[\dot{x}_d - (k_1 x_e)\Big/\sqrt{x_e^2 + y_e^2 + C}\right]\sin\psi + \left[\dot{y}_d - (k_2 y_e)\Big/\sqrt{x_e^2 + y_e^2 + C}\right]\cos\psi \end{cases} \tag{24}$$

where α_u is the virtual control law of longitudinal velocity; α_v is the virtual control law of lateral velocity; k_1, k_2 and C are designed parameters. Considering Eq. (23) and (24), the derivative of Eq. (5) could be designed as follows:

$$\begin{cases} \dot{x}_e = -(k_1 x_e)\Big/\sqrt{x_e^2 + y_e^2 + C} + (u - \alpha_u)\cos\psi + (v - \alpha_v)\sin\psi \\ \dot{y}_e = -(k_2 y_e)\Big/\sqrt{x_e^2 + y_e^2 + C} - (u - \alpha_u)\sin\psi + (v - \alpha_v)\cos\psi \end{cases} \tag{25}$$

Through the above theoretical derivation, the trajectory tracking problem is transformed into designing appropriate longitudinal thrust and yaw moment. The detailed derivation is shown as follows:

4.1.1. The design of the longitudinal thrust

According to Eq. (24), the virtual longitudinal velocity error variable of the USV could be defined as follows:

$$u_e = u - \alpha_u \tag{26}$$

After defining u_e, the sliding surface s_1 could be designed as follows:

$$s_1 = u_e + \lambda_1 \int_0^t u_e(\tau_u)d\tau_u \tag{27}$$

where λ_1 is a designed constant and $\lambda_1 > 0$. The derivative of Eq. (27) could be obtained as:

$$\begin{aligned} \dot{s}_1 &= \dot{u} - \dot{\alpha}_u + \lambda_1 u_e \\ &= \widehat{m}_{22}vr/\widehat{m}_{11} - \widehat{d}_{11}u/\widehat{m}_{11} + d_u/\widehat{m}_{11} + \tau_u/\widehat{m}_{11} - \dot{\alpha}_u + \lambda_1 u_e \end{aligned} \tag{28}$$

The derivative term $\dot{\alpha}_{ui}$ appears in Eq. (28). The resulting complex expanded form would complicate the sliding mode controller design if directly differentiated. The bio-inspired model, first applied in the biological field in paper [31]), was a dynamic model proposed by two scholars, Hodgkin and Huxley, who described the voltage characteristics of biological cell membranes.

To avoid the complex mathematical operations arising from the derivation of the above virtual velocity control laws α_u, α_v, the following bio-heuristic model is adopted:

$$C_m dV_m \Big/ dt = -(E_p + V_m)g_p + (E_{Na} - V_m)g_{Na} - (E_K + V_m)g_K \tag{29}$$

where C_m represents the capacitance; V_m represents the voltage; E_p, E_K and E_{Na} represent the relevant saturation potential, the potassium ion potential inside the cell membrane, and the sodium ion potential,

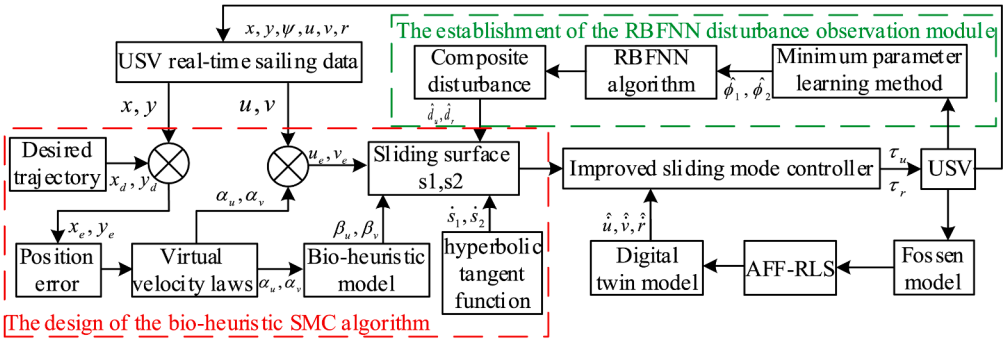


Fig. 3. The controller design process based on digital twin model.

respectively. It is assumed that $C_m = 1$, $x_i = E_p + V_m$, $B = g_p$, $C = E_p + E_{Na}$, $D = E_K - E_p$, $f(x_i) = g_{Na}$, $g(x_i) = g_K$.

Based on the aforementioned bio-heuristic model in Eq. (29), the action potential function could be designed as follows:

$$\dot{x}_i = -Bx_i + (C - x_i)f(x_i) - (D + x_i)g(x_i) \tag{30}$$

where x_i represents the action potential of the *ith* neuron; B, C and D are relevant positive constants representing the decay rate of the model, the upper and lower bounds of x_i, respectively. $f(x_i)$ and $g(x_i)$ represent the external excitatory and inhibitory inputs, which could be described as follows:

$$f(x_i) = \begin{cases} x_i, x_i \geq 0 \\ 0, x_i < 0 \end{cases}; g(x_i) = \begin{cases} 0, x_i > 0 \\ -x_i, x_i \leq 0 \end{cases} \tag{31}$$

It can be seen from Eq. (30) that the potential quantity a is continuous and smooth, and keeps changing within the interval $[-D, B]$ along with the changes of $f(x_i)$,$g(x_i)$. According to Eq. (30) and (31), the approaching function β_u for the virtual longitudinal velocity control law α_u is designed as follows:

$$\dot{\beta}_u = -B_1\beta_u + (C_1 - \beta_u)f(\alpha_u) - (D_1 + \beta_u)g(\alpha_u) \tag{32}$$

Simultaneously, the exponential sliding mode control law in the longitudinal direction based on the hyperbolic tangent function is designed as follows:

$$\dot{s}_1 = -\varepsilon_1 \tanh(s_1) - \sigma_1 s_1 \tag{33}$$

where ε_1 and σ_1 are designed constant and $\varepsilon_1 > 0, \sigma_1 > 0$.

By combining Eq. (28), (32) and (33), the sliding mode control law for the longitudinal thrust (N) of the underactuated USV can be designed as follows:

$$\tau_u = \hat{m}_{11}[-(\hat{m}_{22}vr/\hat{m}_{11} - \hat{d}_{11}u/\hat{m}_{11} + d_u/\hat{m}_{11}) + \dot{\beta}_u - \lambda_1 u_e - \varepsilon_1 \tanh(s_1) - \sigma_1 s_1] \tag{34}$$

4.1.2. The design of the yaw moment

Similarly, according to Eq. (25), the virtual lateral velocity error variable of the USV could be defined as follows:

$$v_e = v - \alpha_v \tag{35}$$

After defining v_e, the sliding surface s_2 could be designed as:

$$s_2 = \dot{v}_e + \lambda_2 v_e + \lambda_3 \int_0^t v_e dt \tag{36}$$

where λ_2 and λ_3 are designed constant and $\lambda_2 > 0, \lambda_3 > 0$. The derivative of Eq. (36) could be obtained as:

$$\dot{s}_2 = \ddot{v} - \ddot{\alpha}_v + \lambda_2(\dot{v} - \dot{\alpha}_v) + \lambda_3(v - \alpha_v) \tag{37}$$

The derivative term $\dot{\alpha}_{vi}$ appears in Eq.(37). The resulting complex expanded form would complicate the sliding mode controller design if directly differentiated. According to Eq. (24), the derivative of α_v could be designed as follows:

$$\begin{aligned} \dot{\alpha}_v = -r\alpha_u - \left[\ddot{x}_d - k_1\left(w^{-1} - w^{-3}x_e^2\right)\dot{x}_e + k_1 w^{-3}x_e y_e \dot{y}_e\right]\sin(\psi) \\ + \left[\ddot{y}_d - k_2\left(w^{-1} - w^{-3}y_e^2\right)\dot{y}_e + k_2 w^{-3}x_e y_e \dot{x}_e\right]\cos(\psi) \end{aligned} \tag{38}$$

where $w = \sqrt{x_e^2 + y_e^2 + C}$. To simplify Eq. (38), defining the designed variable g in the following form:

$$\begin{aligned} g = -\left[\ddot{x}_d - k_1\left(w^{-1} - w^{-3}x_e^2\right)\dot{x}_e + k_1 w^{-3}x_e y_e \dot{y}_e\right]\sin(\psi) \\ + \left[\ddot{y}_d - k_2\left(w^{-1} - w^{-3}y_e^2\right)\dot{y}_e + k_2 w^{-3}x_e y_e \dot{x}_e\right]\cos(\psi) \end{aligned} \tag{39}$$

Then Eq. (38) could be rewritten in the following form:

$$\dot{\alpha}_v = -r\alpha_u + g \tag{40}$$

The derivative of Eq. (40) could be expressed as:

$$\ddot{\alpha}_v = -\dot{r}\alpha_u - r\dot{\alpha}_u + \dot{g} \tag{41}$$

Considering Eq. (4), (37) and (41), the derivative of Eq. (36) could be reconstructed in the following form:

$$\begin{aligned} \dot{s}_2 &= \ddot{v} - (-\dot{r}\alpha_u - r\dot{\alpha}_u + \dot{g}) + \lambda_2(\dot{v} - \dot{\alpha}_v) + \lambda_3(v - \alpha_v) \\ &= \ddot{v} - \{[-(\hat{m}_{11} - \hat{m}_{22})uv/\hat{m}_{22} + \hat{d}_{33}r/\hat{m}_{33} - d_r/\hat{m}_{33}] + \tau_r/\hat{m}_{33}]\alpha_u \\ &\quad -r\dot{\alpha}_u + \dot{g}\} + \lambda_2(\dot{v} - \dot{\alpha}_v) + \lambda_3(v - \alpha_v) \end{aligned} \tag{42}$$

According to Eq. (29), (30) and (31), the approaching function β_v of the virtual lateral velocity control law could be designed as follows:

$$\dot{\beta}_v = -B_2\beta_v + (C_2 - \beta_v)f(\alpha_v) - (D_2 + \beta_v)g(\alpha_v) \tag{43}$$

where B_2, C_2 and D_2 are relevant constants representing the decay rate of

the model, and the upper and lower bounds of β_v, respectively.

Simultaneously, the exponential sliding mode control law in the lateral direction based on the hyperbolic tangent function is designed as follows:

$$\dot{s}_2 = -\varepsilon_2 \tanh(s_2) - \sigma_2 s_2 \tag{44}$$

where ε_2 and σ_2 are designed constant and $\varepsilon_2 > 0, \sigma_2 > 0$.

By considering Eq. (42), (43) and (44), the sliding mode control law for the yaw moment (N·m) of the underactuated USV can be designed as follows:

$$\begin{aligned}\tau_r = \hat{m}_{33}[-\ddot{v} - \lambda_2(\dot{v} - \dot{\beta}_v) - \lambda_3(v - \beta_v) - r\dot{\beta}_u + \dot{g} - \varepsilon_2 \tanh(s_2) \\ -\sigma_2 s_2]/\alpha_u - \hat{m}_{33}[(\hat{m}_{11} - \hat{m}_{22})uv/\hat{m}_{22} - \hat{d}_{33}r/\hat{m}_{33} + d_r/\hat{m}_{33}]\end{aligned} \tag{45}$$

4.2. The establishment of the RBFNN disturbance observation module

To compensate for the influences of composite disturbance on the trajectory tracking of the underactuated USV, which are caused by complex marine environmental disturbances and digital twin modeling errors, an RBFNN algorithm is designed in this paper as shown in Fig. 4 below:

The RBFNN illustrated in Fig.4 is a feedforward neural network consisting of three layers: the input layer, hidden layer, and output layer. Gaussian basis functions are employed, which are non-zero within a limited range of the input space, enabling the approximation of composite disturbances while avoiding the necessity for complex mathematical theoretical analysis and calculations.

The input-output algorithm for the RBFNN can be designed as follows:

$$\hat{m}_{22}vr/\hat{m}_{11} - \hat{d}_{11}u/\hat{m}_{11} + d_u/\hat{m}_{11} = \boldsymbol{W}_u^{*T}\boldsymbol{h}(\boldsymbol{z}) + \delta_1 \tag{46}$$

where $\boldsymbol{z}$ is the input variable of the RBFNN, $\boldsymbol{z} = [u, v, r]^T$; δ_1 is the approximation error of the neural network and $\delta_1 < \delta_U$; δ_U is the upper bound of the approximation error of the neural network and $\delta_U > 0$. $\boldsymbol{W}_u^*$ represent the ideal weight vector from the hidden layer to the output layer, $\boldsymbol{W}_u^* = \left[w_{u,1}^*, w_{u,2}^*, \cdots, w_{u,n}^*\right]^T \in \boldsymbol{R}^{n\times 1}$. $\boldsymbol{h}(\boldsymbol{z})$ is the radial basis function vector of the neural network, $\boldsymbol{h}(\boldsymbol{z}) = [h_1(\boldsymbol{z}), h_2(\boldsymbol{z}), \cdots, h_n(\boldsymbol{z})]^T$. The specific expressions for the components of the radial basis function vector in the neural network could be designed as follows:

$$\boldsymbol{h}_k(\boldsymbol{z}) = \exp\left[-\| \boldsymbol{z} - \boldsymbol{c}_k \|^2 / 2\boldsymbol{b}_k^2\right], (k = 1, 2, \cdots, n) \tag{47}$$

where $\boldsymbol{b}_k$ is the width of the Gaussian basis function; The subscript k represents the k-th node in the hidden layer of the neural network; c_k is the vector value of the center point of the Gaussian basis function, $c_k = [c_1, c_2, \cdots, c_m]^T \in \boldsymbol{R}^m$.

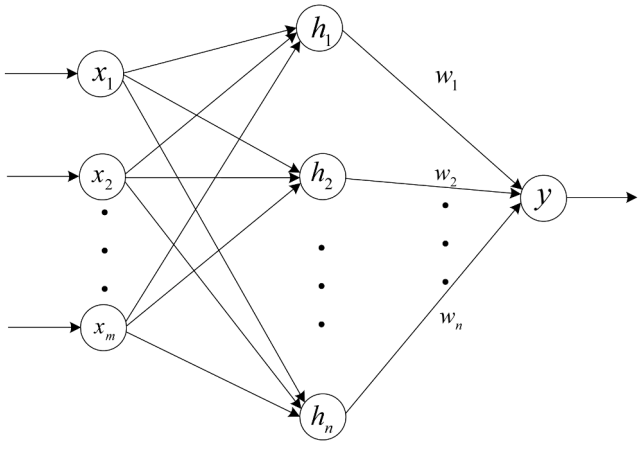


Fig. 4. Structure of RBFNN.

Assuming the estimated value of the ideal weight is $\widehat{\boldsymbol{W}}_u^*$, the estimation function can be further designed as follows:

$$\left(\hat{m}_{22}vr/\hat{m}_{11} - \hat{d}_{11}u/\hat{m}_{11} + d_u/\hat{m}_{11}\right)\widehat{} = \widehat{\boldsymbol{W}}_u^*\boldsymbol{h}(\boldsymbol{z}) \tag{48}$$

Then, a novel sliding mode control law in the longitudinal direction can be designed as follows:

$$\tau_u = \hat{m}_{11}\left[-\widehat{\boldsymbol{W}}_u^*\boldsymbol{h}(\boldsymbol{z}) + \dot{\beta}_u - \lambda_1 u_e - \varepsilon_1 \tanh(s_1) - \sigma_1 s_1\right] \tag{49}$$

The minimum parameter learning method is adopted to let $\phi_1 = \| \boldsymbol{W}_u^* \|^2$, and $\hat{\phi}_1$ is the estimated values of ϕ_1. The estimation error is defined as follows:

$$\tilde{\phi}_1 = \phi_1 - \hat{\phi}_1 \tag{50}$$

According to Eq. (50), the parameter adaptive law is designed as follows:

$$\dot{\hat{\phi}}_1 = \zeta_1\left(s_1^2\boldsymbol{h}^T\boldsymbol{h}/2 - \gamma_1\hat{\phi}_1\right) \tag{51}$$

where ζ_1 and γ_1 are designed variable, $\zeta_1 > 0$, $\gamma_1 > 0$.

Combining Eq. (49) and (51), the longitudinal thrust control law for the underactuated USV is represented in the following form:

$$\tau_u = \hat{m}_{11}\left[-s_1\hat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 + \dot{\beta}_u - \lambda_1 u_e - \varepsilon_1 \tanh(s_1) - \sigma_1 s_1\right] \tag{52}$$

Similarly, the input-output algorithm for the RBFNN in the yaw direction could be designed as follows:

$$(\hat{m}_{11} - \hat{m}_{22})uv/\hat{m}_{22} - \hat{d}_{33}r/\hat{m}_{33} + d_r/\hat{m}_{33} = \boldsymbol{W}_r^{*T}\boldsymbol{h}(\boldsymbol{z}) + \delta_2 \tag{53}$$

where δ_2 is the approximation error of the neural network and $\delta_2 < \delta_R$; δ_R is the upper bound of the approximation error of the neural network and $\delta_R > 0$. $\boldsymbol{W}_r^*$ represent the ideal weight vector from the hidden layer to the output layer, $\boldsymbol{W}_r^* = \left[w_{r,1}^*, w_{r,2}^*, \cdots, w_{r,n}^*\right]^T \in \boldsymbol{R}^{n\times 1}$.

Assuming the estimated value of the ideal weight is $\widehat{\boldsymbol{W}}_r^*$, the estimation function can be further designed as follows:

$$\left[(\hat{m}_{11} - \hat{m}_{22})uv/\hat{m}_{22} - \hat{d}_{33}r/\hat{m}_{33} + d_r/\hat{m}_{33}\right]\widehat{} = \widehat{\boldsymbol{W}}_r^*\boldsymbol{h}(\boldsymbol{z}) \tag{54}$$

Thus, a novel sliding mode control law in the lateral direction can be designed as follows:

$$\begin{aligned}\tau_r = \hat{m}_{33}[-\ddot{v} - \lambda_2(\dot{v} - \dot{\beta}_v) - \lambda_3(v - \beta_v) - r\dot{\beta}_u + \dot{g} \\ -\varepsilon_2 \tanh(s_2) - \sigma_2 s_2]/\alpha_u - \hat{m}_{33}\widehat{\boldsymbol{W}}_r^*\boldsymbol{h}(\boldsymbol{z})\end{aligned} \tag{55}$$

The minimum parameter learning method is adopted to let $\phi_2 = \| \boldsymbol{W}_r^* \|^2$, and $\hat{\phi}_2$ is the estimated values of ϕ_2. The estimation error is defined as follows:

$$\tilde{\phi}_2 = \phi_2 - \hat{\phi}_2 \tag{56}$$

According to Eq. (56), the parameter adaptive law is designed as follows:

$$\dot{\hat{\phi}}_2 = \zeta_2\left(s_2^2\boldsymbol{h}^T\boldsymbol{h}/2 - \gamma_2\hat{\phi}_2\right) \tag{57}$$

where ζ_2 and γ_2 are designed variable, $\zeta_2 > 0$, $\gamma_2 > 0$.

Combining Eq. (56) and (57), the lateral moment control law for the underactuated USV is represented in the following form:

$$\begin{aligned}\tau_r = \hat{m}_{33}[-\ddot{v} - \lambda_2(\dot{v} - \dot{\beta}_v) - \lambda_3(v - \beta_v) - r\dot{\beta}_u \\ +\dot{g} - \varepsilon_2 \tanh(s_2) - \sigma_2 s_2]/\alpha_u - \hat{m}_{33}s_2\hat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2\end{aligned} \tag{58}$$

From Eq. (52) and (58), it can be seen that the control laws of the sliding mode controller designed based on the digital twin model are expressed as follows:

$$\begin{cases} \tau_u = \widehat{m}_{11}\left[-s_1\widehat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 + \dot{\beta}_u - \lambda_1 u_e - \varepsilon_1\tanh(s_1) - \sigma_1 s_1\right] \\ \tau_r = \widehat{m}_{33}[-\ddot{\nu} - \lambda_2(\dot{\nu} - \dot{\beta}_v) - \lambda_3(\nu - \beta_v) - r\dot{\beta}_u \\ \quad +\dot{g} - \varepsilon_2\tanh(s_2) - \sigma_2 s_2]/\alpha_u - \widehat{m}_{33}s_2\widehat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 \end{cases} \tag{59}$$

5. Stability proof

To analyze the stability of the trajectory tracking controller in Eq. (59), based on the above derivation and design, the following theorem can be obtained:

Theorem 1: . The trajectory tracking controller in Eq. (59) designed in this paper ensures that the trajectory tracking errors x_e, y_e defined in Eq. (5) converge to zero.

Proof : The proof proceeds in two key steps: In Step 1, the convergence of the actual velocities to their designed virtual velocities ($u \to \alpha_u$, $v \to \alpha_v$) will be validated; In Step 2, a unified stability analysis for the sliding surface errors, bio-heuristic model approximation errors and the neural network-based disturbance observation errors will be conducted.

Step 1: stability analysis of the virtual control laws

Assuming the virtual velocity errors in longitudinal and lateral directions could be stabilized, then $u - \alpha_u \to 0, v - \alpha_v \to 0$. Eq. (25) could be reconstructed as follows:

$$\begin{cases} \dot{x}_e = -k_1 x_e \Big/ \sqrt{x_e^2 + y_e^2 + C} \\ \dot{y}_e = -k_2 y_e \Big/ \sqrt{x_e^2 + y_e^2 + C} \end{cases} \tag{60}$$

Then the Lyapunov function of the position errors could be designed as follows:

$$V_1 = x_e^2/2 + y_e^2/2 \tag{61}$$

The derivative of Lyapunov function V_1 could be obtained as:

$$\dot{V}_1 = x_e\dot{x}_e + y_e\dot{y}_e = -\left(k_1 x_e^2 + k_2 y_e^2\right) \Big/ \sqrt{x_e^2 + y_e^2 + C} \tag{62}$$

Considering Eq. (62) and Lyapunov stability theory, it is known that if the virtual velocity errors are stabilized, the trajectory tracking error will converge to a stable state.

Step 2: unified stability analysis

To further prove that the underactuated USV's trajectory tracking errors has global uniform stability, the following Lyapunov function is designed:

$$V_2 = s_1^2/2 + s_2^2/2 + y_u^2/2 + y_v^2/2 + \widetilde{\phi}_1^2/2\zeta_1 + \widetilde{\phi}_2^2/2\zeta_2 \tag{63}$$

where y_u and y_v are bio-heuristic model approximation errors. The derivative of Lyapunov function V_2 could be obtained as:

$$\dot{V}_2 = s_1\dot{s}_1 + s_2\dot{s}_2 + y_u\dot{y}_u + y_v\dot{y}_v + \widetilde{\phi}_1\dot{\widetilde{\phi}}_1/\zeta_1 + \widetilde{\phi}_2\dot{\widetilde{\phi}}_2/\zeta_2 \tag{64}$$

According to Eq. (28) and (37), the derivative of sliding surfaces is obtained as:

$$\begin{cases} \dot{s}_1 = W_u^{*T}\boldsymbol{h}(\boldsymbol{z}) + \delta_1 - s_1\widehat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 - \varepsilon_1\tanh(s_1) - \sigma_1 s_1 \\ \dot{s}_2 = W_r^{*}\boldsymbol{h}(\boldsymbol{z}) + \delta_2 - s_2\widehat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 - \varepsilon_2\tanh(s_2) - \sigma_2 s_2 \end{cases} \tag{65}$$

By substituting Eq. (65) into Eq. (64), the derivative of Lyapunov function V_2 in Eq. (64) could be reconstructed as:

$$\begin{aligned} \dot{V}_2 &= s_1\left(W_u^{*}\boldsymbol{h}(\boldsymbol{z}) + \delta_1 - s_1\widehat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 - \varepsilon_1\tanh(s_1) - \sigma_1 s_1\right) + s_2(W_r^{*}\boldsymbol{h}(\boldsymbol{z}) + \delta_2 - s_2\widehat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 \\ &\quad -\varepsilon_2\tanh(s_2) - \sigma_2 s_2) + y_u(\dot{\beta}_u - \dot{\alpha}_u) + y_v(\dot{\beta}_v - \dot{\alpha}_v) + \widetilde{\phi}_1(\dot{\phi}_1 - \dot{\widehat{\phi}}_1)/\zeta_1 + \widetilde{\phi}_2(\dot{\phi}_2 - \dot{\widehat{\phi}}_2)/\zeta_2 \\ &\le s_1\left(W_u^{*}\boldsymbol{h}(\boldsymbol{z}) + \delta_1 - s_1\widehat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 - \sigma_1 s_1\right) - \varepsilon_1|s_1| + s_2(W_r^{*}\boldsymbol{h}(\boldsymbol{z}) + \delta_2 - s_2\widehat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 \\ &\quad -\sigma_2 s_2) - \varepsilon_2|s_2| + y_u(\dot{\beta}_u - \dot{\alpha}_u) + y_v(\dot{\beta}_v - \dot{\alpha}_v) - \widetilde{\phi}_1\dot{\widehat{\phi}}_1/\zeta_1 - \widetilde{\phi}_2\dot{\widehat{\phi}}_2/\zeta_2 \end{aligned} \tag{66}$$

Considering the principle of inequality, the relationship can be obtained as follows:

$$\begin{cases} s_1\delta_1 \le {s_1}^2/2 + {\delta_1}^2/2 \le {s_1}^2/2 + {\delta_U}^2/2 \\ s_2\delta_2 \le {s_2}^2/2 + {\delta_2}^2/2 \le {s_2}^2/2 + {\delta_R}^2/2 \\ s_1 W_u^{*}\boldsymbol{h}(\boldsymbol{z}) \le \left({s_1}^2\| W_u^{*}\|^2\boldsymbol{h}(\boldsymbol{z})^2 + 1\right)/2 = \left({s_1}^2\phi_1\boldsymbol{h}^T\boldsymbol{h} + 1\right)/2 \\ s_2 W_r^{*}\boldsymbol{h}(\boldsymbol{z}) \le \left({s_2}^2\| W_r^{*}\|^2\boldsymbol{h}(\boldsymbol{z})^2 + 1\right)/2 = \left({s_2}^2\phi_2\boldsymbol{h}^T\boldsymbol{h} + 1\right)/2 \end{cases} \tag{67}$$

Bringing Eq. (67) into Eq. (66), Eq. (66) could be expressed as follows:

$$\begin{aligned} \dot{V}_2 &\le \left({s_1}^2\phi_1\boldsymbol{h}^T\boldsymbol{h} + 1\right)/2 + {s_1}^2/2 + {\delta_U}^2/2 - \left({s_1}^2\widehat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}\right)/2 - \sigma_1{s_1}^2 \\ &\quad -\varepsilon_1|s_1| + \left({s_2}^2\phi_2\boldsymbol{h}^T\boldsymbol{h} + 1\right)/2 + {s_2}^2/2 + {\delta_R}^2/2 - \left({s_2}^2\widehat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}\right)/2 \\ &\quad -\sigma_2{s_2}^2 - \varepsilon_2|s_2| + y_u(\dot{\beta}_u - \dot{\alpha}_u) + y_v(\dot{\beta}_v - \dot{\alpha}_v) - \widetilde{\phi}_1\dot{\widehat{\phi}}_1/\zeta_1 - \widetilde{\phi}_2\dot{\widehat{\phi}}_2/\zeta_2 \end{aligned} \tag{68}$$

where $\dot{\beta}_u$ and $\dot{\beta}_v$ could be written in the following form based on the bio-heuristic model in Eq. (30):

$$\begin{cases} \dot{\beta}_u = -B_1\beta_u + (C_1 - \beta_u)f(\alpha_u) - (D_1 + \beta_u)g(\alpha_u) \\ = \beta_u[-B_1 - f(\alpha_u) - g(\alpha_u)] + C_1 f(\alpha_u) - D_1 g(\alpha_u) \\ \dot{\beta}_v = -B_2\beta_v + (C_2 - \beta_v)f(\alpha_v) - (D_2 + \beta_v)g(\alpha_v) \\ = \beta_v[-B_2 - f(\alpha_v) - g(\alpha_v)] + C_2 f(\alpha_v) - D_2 g(\alpha_v) \end{cases} \tag{69}$$

Then Eq. (68) could be expressed as follows:

$$\begin{aligned} \dot{V}_2 &\le \left({s_1}^2\phi_1\boldsymbol{h}^T\boldsymbol{h} + 1\right)/2 + {s_1}^2/2 + {\delta_U}^2/2 - {s_1}^2\widehat{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 - \sigma_1{s_1}^2 - \varepsilon_1|s_1| \\ &\quad +\left({s_2}^2\phi_2\boldsymbol{h}^T\boldsymbol{h} + 1\right)/2 + {s_2}^2/2 + {\delta_R}^2/2 - {s_2}^2\widehat{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 - \sigma_2{s_2}^2 - \varepsilon_2|s_2| \\ &\quad +y_u[-\beta_u(B_1 + f(\alpha_u) + g(\alpha_u)) + C_1 f(\alpha_u) - D_1 g(\alpha_u) - \dot{\alpha}_u] - \widetilde{\phi}_1\dot{\widehat{\phi}}_1/\zeta_1 \\ &\quad +y_v[-\beta_v(B_2 + f(\alpha_v) + g(\alpha_v)) + C_2 f(\alpha_v) - D_2 g(\alpha_v) - \dot{\alpha}_v] - \widetilde{\phi}_2\dot{\widehat{\phi}}_2/\zeta_2 \end{aligned} \tag{70}$$

According to Eq. (31), when $x \ge 0$, it is obvious that $f(x) = x, g(x) = 0$; When $x < 0$, $f(x) = 0, g(x) = -x$; Assuming that $C = D$, then the expanded expressions of B_u and B_v could be designed as follows:

$$\begin{cases} B_u = B_1 + f(\alpha_u) + g(\alpha_u) > 0 \\ B_r = B_2 + f(\alpha_r) + g(\alpha_r) > 0 \end{cases} \tag{71}$$

Bringing Eq. (71) into (70), the relationship could be expressed as follows:

$$\begin{aligned} \dot{V}_2 &\le {s_1}^2\widetilde{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 + {s_2}^2\widetilde{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 - {s_1}^2(2\sigma_1 - 1)/2 - {s_2}^2(2\sigma_2 - 1)/2 + {\delta_U}^2/2 \\ &\quad +{\delta_R}^2/2 - \varepsilon_1|s_1| - \varepsilon_2|s_2| + 1 + y_u[-\beta_u B_u + C_1(f(\alpha_u) - g(\alpha_u)) - \dot{\alpha}_u] \\ &\quad +y_v[-\beta_v B_v + C_2(f(\alpha_v) - g(\alpha_v)) - \dot{\alpha}_v] - \widetilde{\phi}_1\dot{\widehat{\phi}}_1/\zeta_1 - \widetilde{\phi}_2\dot{\widehat{\phi}}_2/\zeta_2 \end{aligned} \tag{72}$$

Assuming that $B_u = C_1, B_v = C_2, \alpha_u = f(\alpha_u) - g(\alpha_u), y_u = \beta_u - \alpha_u$, and $y_v = \beta_v - \alpha_v$, then Eq. (72) could be expressed as follows:

$$\dot{V}_2 \le {s_1}^2\widetilde{\phi}_1\boldsymbol{h}^T\boldsymbol{h}/2 + {s_2}^2\widetilde{\phi}_2\boldsymbol{h}^T\boldsymbol{h}/2 - {s_1}^2(2\sigma_1-1)/2 - {s_2}^2(2\sigma_2-1)/2 + {\delta_U}^2/2 + {\delta_R}^2/2 - \varepsilon_1|s_1| - \varepsilon_2|s_2| + 1 - {y_u}^2B_u - y_u\dot{\alpha}_u - {y_v}^2B_v - y_v\dot{\alpha}_v - \widetilde{\phi}_1\dot{\widehat{\phi}}_1/\zeta_1 - \widetilde{\phi}_2\dot{\widehat{\phi}}_2/\zeta_2 \tag{73}$$

Considering the parameter adaptive laws in Eq. (51) and (57), Eq. (73) could be rewritten as:

$$\dot{V}_2 \le -{s_1}^2(2\sigma_1-1)/2 - {s_2}^2(2\sigma_2-1)/2 + {\delta_U}^2/2 + {\delta_R}^2/2 + 1 - {y_u}^2B_u - y_u\dot{\alpha}_u - {y_v}^2B_v - y_v\dot{\alpha}_v + \gamma_1\widehat{\phi}_1\widetilde{\phi}_1 + \gamma_2\widehat{\phi}_2\widetilde{\phi}_2 \tag{74}$$

where ε_1 and ε_2 are designed constants and $\varepsilon_1 > 0, \varepsilon_2 > 0$, $-\varepsilon_1|s_1| - \varepsilon_2|s_2| \le 0$. It can be derived from the properties of inequalities that:

$$\begin{cases} y_u\dot{\alpha}_u \le \kappa_u y_u^2/2 + \dot{\alpha}_u^2/2\kappa_u \le \kappa_u y_u^2/2 + U_{\max}^2/2\kappa_u \\ y_v\dot{\alpha}_v \le \kappa_v y_v^2/2 + \dot{\alpha}_v^2/2\kappa_v \le \kappa_v y_v^2/2 + V_{\max}^2/2\kappa_v \end{cases} \tag{75}$$

where $\dot{\alpha}_u$ and $\dot{\alpha}_v$ are bounded, $U_{\max}$ is the maximum value of $\dot{\alpha}_u$, $V_{\max}$ is the maximum value of $\dot{\alpha}_v$. By the properties of inequalities, $\widetilde{\phi}_1\widehat{\phi}_1$ and $\widetilde{\phi}_2\widehat{\phi}_2$ could be expressed in the following form:

$$\begin{cases} 2\widetilde{\phi}_1\widehat{\phi}_1 = (\phi_1 - \widehat{\phi}_1)\widehat{\phi}_1 + \widetilde{\phi}_1(\phi_1 - \widetilde{\phi}_1) \\ \quad = \phi_1^2 - \widehat{\phi}_1^2 - \widetilde{\phi}_1^2 \le \phi_1^2 - \widetilde{\phi}_1^2 \\ 2\widetilde{\phi}_2\widehat{\phi}_2 = (\phi_2 - \widehat{\phi}_2)\widehat{\phi}_2 + \widetilde{\phi}_2(\phi_2 - \widetilde{\phi}_2) \\ \quad = \phi_2^2 - \widehat{\phi}_2^2 - \widetilde{\phi}_2^2 \le \phi_2^2 - \widetilde{\phi}_2^2 \end{cases} \tag{76}$$

Bringing Eq. (75) and (76) into Eq. (74), Eq. (74) could be obtained that:

$$\begin{aligned} \dot{V}_2 &\le -{s_1}^2(2\sigma_1-1)/2 - {s_2}^2(2\sigma_2-1)/2 - {y_u}^2B_u + \kappa_u y_u^2/2 + U_{\max}^2/2\kappa_u - {y_v}^2B_v \\ &+\kappa_v y_v^2/2 + V_{\max}^2/2\kappa_v - \gamma_1\widetilde{\phi}_1^2/2 - \gamma_2\widetilde{\phi}_2^2/2 + {\delta_U}^2/2 + {\delta_R}^2/2 + 1 + \gamma_1\phi_1^2/2 + \gamma_2\phi_2^2/2 \\ &= -{s_1}^2(2\sigma_1-1)/2 - {s_2}^2(2\sigma_2-1)/2 - {y_u}^2(B_u - \kappa_u/2) - {y_v}^2(B_v - \kappa_v/2) - \widetilde{\phi}_1^2\gamma_1/2 \\ &\quad -\widetilde{\phi}_2^2\gamma_2/2 + U_{\max}^2/2\kappa_u + V_{\max}^2/2\kappa_v + {\delta_U}^2/2 + {\delta_R}^2/2 + 1 + \gamma_1\phi_1^2/2 + \gamma_2\phi_2^2/2 \end{aligned} \tag{77}$$

In order to make $\dot{V}_2 \le 0$, adjusting parameters $\sigma_1 \ge 1/2, \sigma_2 \ge 1/2$. Then Eq. (77) could be simplified into the following forms:

$$\dot{V}_2 \le -\mu V_2 + C_3 \tag{78}$$

where$\mu = \min\{(2\sigma_1 - 1)/2, (2\sigma_2 - 1)/2, B_u - \kappa_u/2, B_v - \kappa_v/2, \gamma_1/2, \gamma_2/2\} > 0$; and$C_3 = U_{\max}^2/2\kappa_u + V_{\max}^2/2\kappa_v + {\delta_U}^2/2 + {\delta_R}^2/2 + 1 + \gamma_1\phi_1^2/2 + \gamma_2\phi_2^2/2$. Assuming $k_1 = 1$, $k_2 = 1$, $k_3 = \mu$ and $\rho = C_3$, according to literature [32], the Eq. (78) can be shown as:

$$V_2(t) \le [V_2(0) - C_3/\mu]e^{-\mu t} + C_3/\mu \tag{79}$$

According to Eq. (79), it could be concluded that the states V_2 is ultimately bounded by C_3/μ. Since V_2 is a quadratic function of the state $\xi = [s_1, s_2, y_u, y_v, \widetilde{\phi}_1, \widetilde{\phi}_2]^T$, this implies that all these error signals are ultimately uniformly bounded and converge to the compact set:$\Omega = \left\{\xi \in R^6 \mid \| \xi \| \le \sqrt{2C_3/\mu}\right\}$. This means the system states are ultimately uniformly bounded, and $[V_2(0) - C_3/\mu]e^{-\mu t}$ approaches zero. By combining Eq. (78), it is proven that $\dot{V}_2 \le 0$, which means that the error values $s_1, s_2, y_u, y_v, \widetilde{\phi}_1, \widetilde{\phi}_2$ in u_e, v_e will converge stably to zero. Furthermore, it is also shown that u_e, v_e themselves will stably converge to zero. Therefore, by combining Eq. (62) in Step 1 and Lyapunov's stability theory, it can be proved that the trajectory tracking errors x_e, y_e in Eq. (5) can stably converge to zero.

In summary, through the above proof process, Theorem 1 is comprehensively proven. Therefore, the trajectory tracking controller designed in this paper could achieve that the tracking errors converges to zero.

6. Computer simulation

To verify the effectiveness of the algorithm proposed in this paper, several comparative simulation experiments are conducted under three working conditions. Four methods are considered in the simulation experiments: Method 1: the bio-heuristic sliding mode controller designed based on the traditional model with fixed parameters; Method 2: the bio-heuristic sliding mode controller designed based on the digital twin model, integrated with RLS algorithm; Method 3: the bio-heuristic sliding mode controller designed based on the digital twin model, integrated with AFF-RLS algorithm; Method 4: the bio-heuristic sliding mode controller designed based on digital twin model, integrated with AFF-RLS algorithm and RBFNN-based disturbance compensation module. The hardware platform is 2.59-GHz CPU and 8-GB memory and the software platform is OCTAVE 9.3.0.

6.1. Parameter settings

According to the literature [33], the model parameters of the USV Fossen model are given in Table 1.

The actuator constraints considered in the control process are as follows:

$$\begin{aligned} &\boldsymbol{\tau}_{\min} = [0(N), -30(N{\cdot}m)]^T, \boldsymbol{\tau}_{\max} = [100(N), 30(N{\cdot}m)]^T; \\ &\Delta\boldsymbol{\tau}_{\min} = [-20(N), -10(N{\cdot}m)]^T, \Delta\boldsymbol{\tau}_{\max} = [20(N), 10(N{\cdot}m)]^T; \end{aligned} \tag{80}$$

where $\boldsymbol{\tau}$ represent the output force (N) and moment (N·m); $\Delta\boldsymbol{\tau}$ denote the change rate of the output force and moment. Additionally, the marine environmental disturbances defined as follows:

Table 1
model parameters.

parameter	value(unit)	parameter	value(unit)	parameter	value(unit)
m_{11}	$215kg$	m_{22}	$265kg$	m_{33}	$80kg{\cdot}m^2$
d_{11}	$70kg/s$	d_{22}	$100kg/s$	d_{33}	$50kg{\cdot}m^2/s$

Table 2
The initial conditions and control parameters of the USV (Case I).

parameter	value (unit)	parameter	value (unit)	parameter	value(unit)
T_1	$150(s)$	dt	$0.1(s)$	*state*	$[-5, -5, \pi/4, 0, 0, 0]$
k	30	C	2	k_1	7
k_2	2.1	λ_1	0.6	λ_2	4
λ_3	0.001	ζ_1	0.01	ζ_2	0.1
ε_1	0.00001	ε_2	0.01	B_1	3
B_2	0.5	C_1	2.5	C_2	0.65
D_1	1	D_2	0.65	σ_1	0.1
σ_2	0.01	γ_1	0.001	γ_2	0.001

Table 3
The initial conditions and control parameters of the USV (Case II).

parameter	value (unit)	parameter	value (unit)	parameter	value(unit)
T_1	150(s)	dt	0.1(s)	$state$	$[-10,-10,\pi/4,0,0,0]$
k	30	C	20	k_1	10
k_2	4	λ_1	1.5	λ_2	1.2
λ_3	0.01	ζ_1	0.01	ζ_2	1
ε_1	0.01	ε_2	1	B_1	3.8
B_2	0.5	C_1	2.85	C_2	0.65
D_1	15	D_2	0.5	σ_1	1
σ_2	0.05	γ_1	0	γ_2	0.01

Table 4
The initial conditions and control parameters of the USV (Case III).

parameter	value (unit)	parameter	value (unit)	parameter	value(unit)
T_1	150(s)	dt	0.1(s)	$state$	$[-5,-5,\pi/2,0.1,0,0]$
k	40	C	0.5	k_1	5
k_2	5	λ_1	3	λ_2	0.3
λ_3	0.01	ζ_1	1	ζ_2	1
ε_1	0.01	ε_2	1	B_1	4
B_2	0.5	C_1	2.5	C_2	0.65
D_1	5	D_2	0.65	σ_1	0.1
σ_2	0.01	γ_1	0.001	γ_2	0.001

$$\mathbf{d}_o = [3\sin(0.1t), 0.1\cos(0.1t), \sin(0.1t)] \tag{81}$$

In addition, all simulations adopt a fixed time step dt=0.1 s for discretized solving, which simultaneously serves as the input update rate for the RLS algorithm, ensuring strict synchronization between state data and recursive computation in Eq.(20).

6.2. Simulation results

In order to directly assess USV's trajectory tracking performance, the position error of the USV is defined as $e_{xy} = \sqrt{(x-x_d)^2 + (y-y_d)^2}$. To stabilize the identification algorithm during its initial stage, the initial model parameters are applied for the first k iterations. Meanwhile, to ensure that proposed algorithms meet the real-time requirements, the total computation time of different algorithms in different working conditions will be compared in Table 5. The simulation results are provided as follows:

Case I. straight-line trajectory tracking

The desired trajectory information is $x_d = t, y_d = t$. The discrete time interval for straight-line reference trajectories is set to dt=0.1 s, consistent with the controller sampling period to avoid additional errors from temporal resolution mismatch. The initial conditions and control parameters of the USV are provided in Table 2.

The simulation results of Case I are provided as follows:

Case II. circular trajectory tracking

The desired trajectory information is $x_d = 30sin(0.05t), y_d = 30 - 30sin(0.05t)$. The discrete time interval for circular reference trajectories is set to dt=0.1 s, consistent with the controller sampling period to avoid additional errors from temporal resolution mismatch. The initial conditions and control parameters of the USV are provided in Table 3.

The simulation results of Case II are provided as follows:

Case III. sinusoidal trajectory tracking

The desired trajectory information is$x_d = t, y_d = 25\sin(0.05t)$. The discrete time interval for sinusoidal reference trajectories is set to dt=0.1 s, consistent with the controller sampling period to avoid additional errors from temporal resolution mismatch. The initial conditions and control parameters of the USV are provided in Table 4.

The simulation results of Case III are provided as follows:

The simulation duration for three different working conditions is 150 seconds, with a time step of 0.1 seconds. The runtime for proposed algorithms is as follows:

Table 5
The total runtime of the proposed algorithms.

Method/Case	Case I	Case II	Case III
RLS	0.189s	0.190s	0.196s
AFF-RLS	0.209s	0.204s	0.215s
SMC	0.131s	0137s	0.139s

According to Table 5, a comparison of the total runtime between RLS algorithm and AFF-RLS algorithm shows that AFF-RLS algorithm requires more computational effort because it dynamically adjusts the forgetting factor, and its total computation time is longer. However, it is evident that even though AFF-RLS algorithm takes more time, its runtime is still significantly less than 0.1 seconds. This demonstrates that both proposed RLS and AFF-RLS algorithms exhibit excellent real-time performance. Moreover, the total runtime demonstrates that the proposed SMC design meets the requirements for real-time performance.

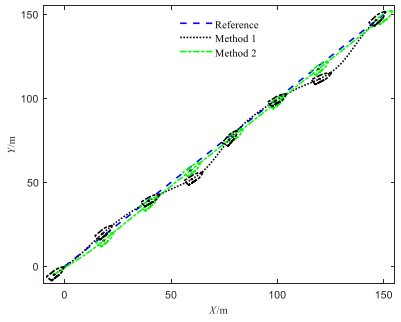
The experimental results presented in this section demonstrate the trajectory tracking control performance of the underactuated USV under complex marine environmental disturbances for each method. As shown in (a),(f) and (g) in Figs. 5, 6, and 7, a comparison is conducted between the desired trajectory and the real trajectory obtained utilizing Method 1 and Method 2 under three different working conditions. The position errors for each method are calculated and compared across the three conditions. It is evident that fixed parameters lead to significant position errors in the trajectory tracking of the USV. In contrast, trajectory tracking performance is significantly improved by applying the RLS algorithm to obtain model parameters. As shown in (f) and (g) in Figs. 5, 6 and 7, Method 2 not only makes the tracking process more stable, but also optimizes the accuracy.

Through analysis of the controller's output and the USV's velocity values, the robustness of the control algorithm proposed in this paper is further demonstrated, as shown in (d) and (e) in Figs. 5, 6, and 7. Notably, compared to Method 2, Method 3 exhibits a smaller initial trajectory tracking error, which converges more quickly to a smaller position error, highlighting the superiority of the AFF-RLS algorithm. Additionally, as shown in (b), (f) and (g) in Figs. 5, 6, and 7, the stability and anti-disturbance capability of Method 3 are enhanced, thanks to the adaptive forgetting factor optimization. This improvement enables the algorithm to more effectively address digital twin modeling errors and marine disturbances, ensuring that the USV maintains its tracking performance even in the presence of composite disturbances.

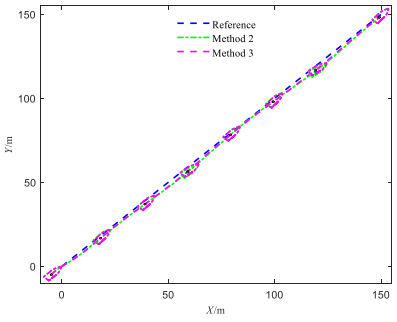
As shown in (c), (f) and (g) in Figs. 5, 6, and 7, illustrate the tracking performance and position errors of the USV under three different working conditions. A comparison between method 4 and method 3 reveals that applying the RBFNN to compensate for composite disturbances significantly enhances the robustness of the control algorithm. By integrating the RBFNN, the control algorithm can effectively respond to the composite disturbance and dynamically adjust the controller's output to minimize their impact. Schematic in (f) and (g) in Figs. 5, 6, and 7, further demonstrates that the use of the RBFNN substantially improves the control algorithm's ability to maintain the USV's position and trajectory accuracy.

6. Conclusion

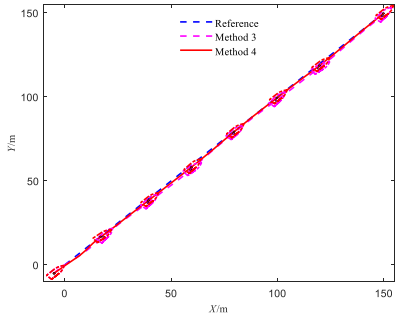
The bio-heuristic SMC algorithm, based on a digital twin model, has



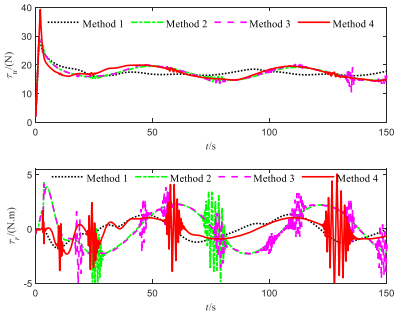
(a) USV's trajectory (Method1,2)



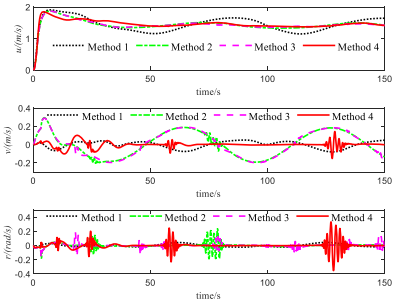
(b) USV's trajectory (Method2,3)



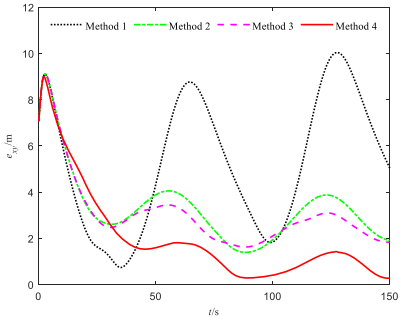
(c) USV's trajectory (Method3,4)



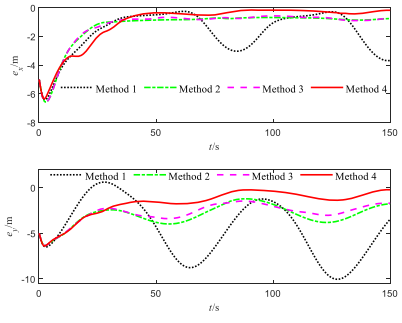
(d) USV control variables (Case I)



(e) USV velocity variables (Case I)

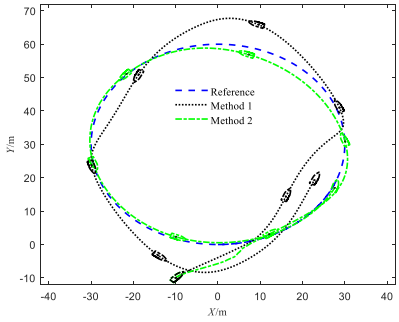


(f) USV's position error (Case I)

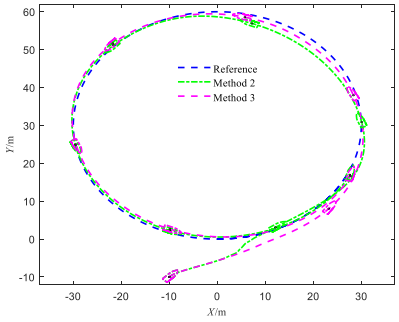


(g) position error in longitudinal and lateral directions

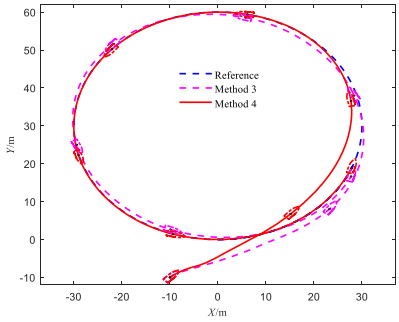
Fig. 5. USV's control performance and position error (Case I).



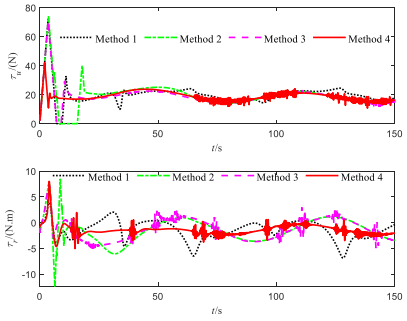
(a) USV's trajectory (Method1,2)



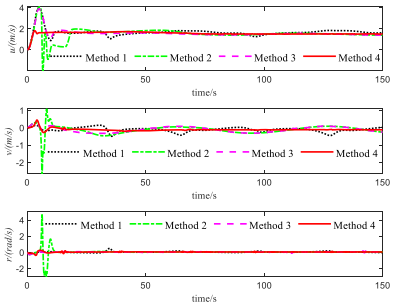
(b) USV's trajectory (Method2,3)



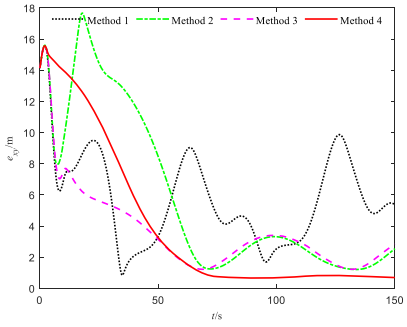
(c) USV's trajectory (Method3,4)



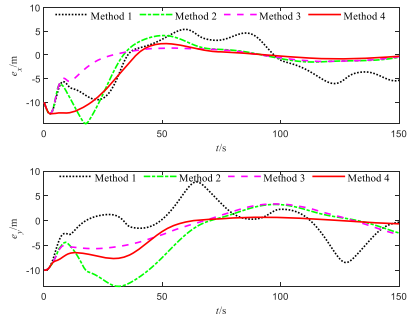
(d) USV control variables (Case II)



(e) USV velocity variables (Case II)

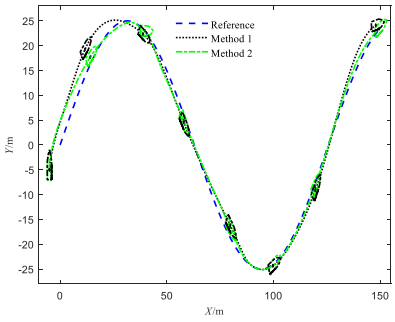


(f) USV's position error (Case II)

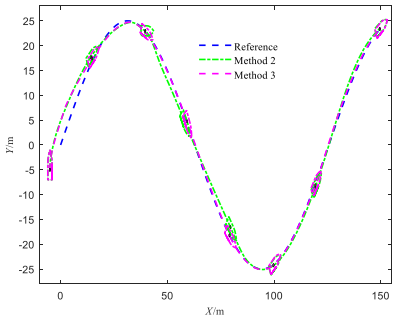


(g) position error in longitudinal and lateral directions

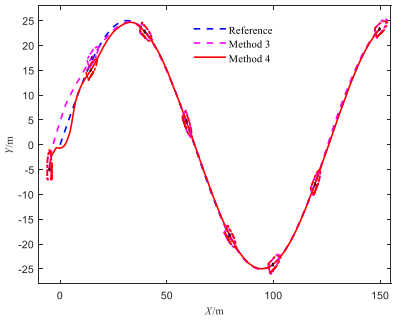
Fig. 6. USV's control performance and position error (Case II).



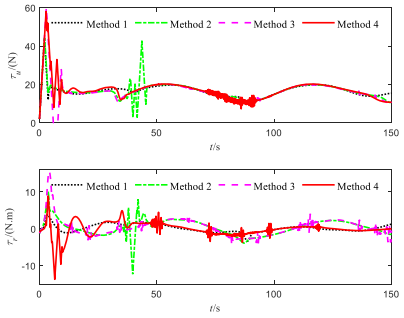
(a) USV's trajectory (Method1,2)



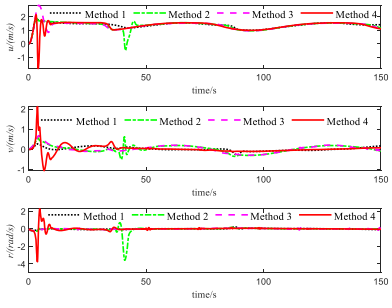
(b) USV's trajectory (Method2,3)



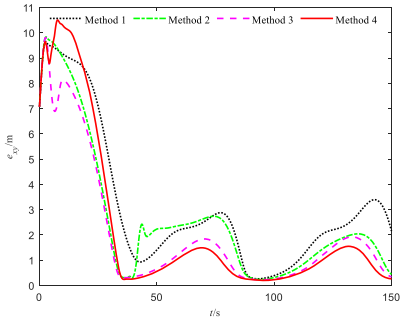
(c) USV's trajectory (Method3,4)



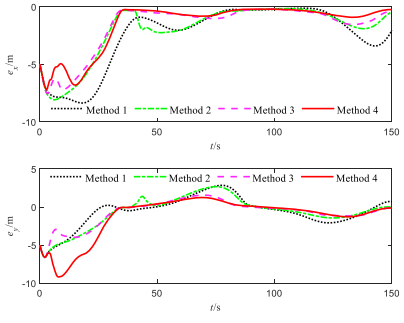
(d) USV control variables (Case III)



(e) USV velocity variables (Case III)



(f) USV's position error (Case III)



(g) position error in longitudinal and lateral directions

Fig. 7. USV's control performance and position error (Case III).

been demonstrated to significantly improve the control performance of the USV, especially under the influence of marine environmental disturbances. By leveraging the digital twin model, the proposed algorithm ensures more accurate tracking of the USV's trajectory in real-time, adapting effectively to varying environmental conditions. The incorporation of an adaptive forgetting factor into the RLS algorithm has enhanced the precision of the digital twin model by dynamically adjusting the model parameters based on real-time state variables and control commands. Additionally, the adoption of a bio-heuristic approximation function to approach the virtual velocity control laws has significantly improved the performance of the sliding mode controller. By replacing the traditional sign function with a hyperbolic tangent function, the USV has eliminated abrupt control changes and jittering, which are common in trajectory tracking control, thereby ensuring smoother and more stable trajectory tracking. Furthermore, the integration of a RBFNN-based compensator has been proven to bolster the algorithm's robustness. The compensator effectively addresses environmental disturbances and digital twin modeling errors, compensating for composite disturbances that would otherwise degrade the tracking performance. The RBFNN-based approach allows for real-time disturbance estimation and correction, further enhancing the controller's ability to maintain stable performance in the face of dynamic and unknown disturbances. In conclusion, the effectiveness and robustness of the proposed digital twin model-based SMC algorithm for USV under complex marine environmental disturbances have been thoroughly validated through a series of extensive simulation experiments in this paper. These experiments have demonstrated that the proposed algorithm not only improves trajectory tracking accuracy but also ensures stability and robustness in a wide range of working conditions.

CRediT authorship contribution statement

Zaopeng Dong: Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Sihang Lu:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Zhihao Hu:** Writing – review & editing, Writing – original draft. **Wangsheng Liu:** Writing – review & editing, Writing – original draft. **Yilun Ding:** Writing – review & editing, Writing – original draft. **Yuanchang Liu:** Writing – review & editing.

Declaration of competing interest

No conflict of interest exits in the submission of this manuscript, and manuscript is approved by all authors for publication. On behalf of my co-authors, I would like to declare that the work described is original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript is enclosed.

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Data availability

No data was used for the research described in the article.

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